

CineMood AI: Personalized Movie Recommendation System

1. Introduction

CineMood AI is an AI-based movie recommendation system that suggests movies based on user ratings, mood, and preferred genre. The project uses collaborative filtering with cosine similarity to recommend movies by finding users with similar movie preferences.

To make the system more creative and interactive, features like mood-based filtering, Movie Twin, Movie Personality, CineScore ranking, explainable recommendations, cold-start recommendations, weekend watchlist, and a Gradio interface were added.

2. Objective

The main objective of this project is to build a movie recommendation system that can suggest suitable movies to users based on their interests and rating patterns.

The project aims to:

- Collect and use a movie ratings dataset.
- Clean and preprocess the dataset using Python.
- Analyze movie ratings and genres.
- Implement collaborative filtering.
- Recommend movies for different users.
- Create visualizations for popular movies and rating trends.
- Build an interactive interface for users.

3. Dataset Used

The MovieLens Latest Small Dataset was used for this project. The main files used were:

- movies.csv
- ratings.csv

The movies dataset contains movie ID, title, and genre details. The ratings dataset contains user ID, movie ID, rating, and timestamp details. Both datasets were merged using the movieid column.

4. Tools and Technologies

The tools and technologies used in this project are:

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Google Colab
- Gradio

Pandas was used for data cleaning and analysis. Matplotlib was used for visualizations. Scikit-learn was used to calculate cosine similarity. Gradio was used to create the interactive web interface.

5. Methodology

First, the movies and ratings datasets were loaded into Google Colab. The datasets were checked for missing values and duplicate records. Then, both datasets were merged using the movieid column.

After preprocessing, exploratory data analysis was performed to understand rating distribution, popular movies, and movie genres.

A user-movie matrix was created where rows represent users and columns represent movies. The values in the matrix represent ratings given by users. Missing values were filled with zero. Cosine similarity was then applied to find similar users. Movies liked by similar users were recommended to the selected user.

6. Recommendation Features

The main features of CineMood AI are:

- User-based collaborative filtering
- Mood-based movie recommendation

- Preferred genre filtering
- Movie Twin feature
- Movie Personality detection
- CineScore ranking
- Explainable recommendation reasons
- Cold-start recommendation for new users
- Weekend watchlist generator
- Interactive Gradio interface

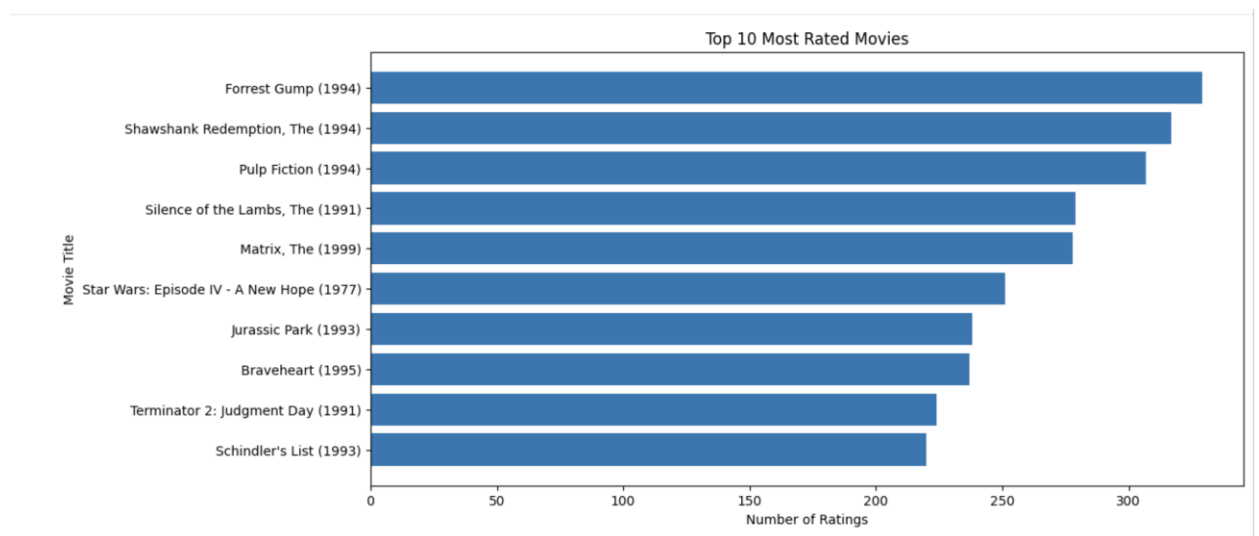
CineScore is a custom score used to rank movies. It combines similar user rating score, average movie rating, and popularity score.

7. Visualizations

The project includes the following visualizations:

- Top 10 most rated movies
- Rating distribution
- Top 10 movie genres
- CineScore comparison chart

1. TOP 10 RATED MOVIES



```

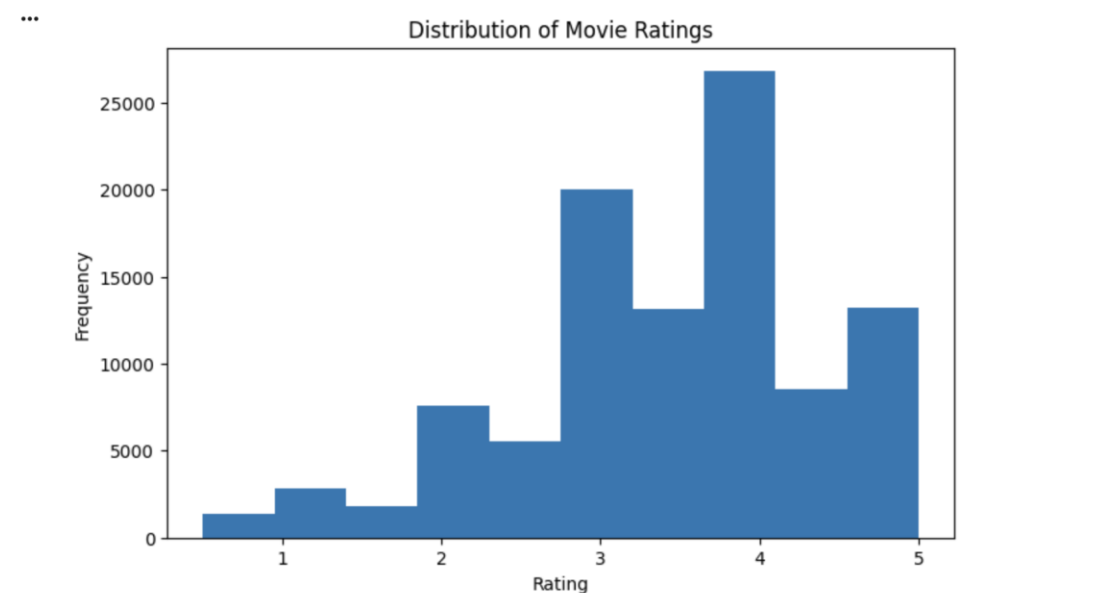
▶ advanced_recs = advanced_cinemood_ai(
    user_id=25,
    mood="adventurous",
    preferred_genre="Sci-Fi",
    num_recommendations=5
)

... 📄 CineMood AI - Advanced Movie Recommendation System
=====
Mood Selected: adventurous
Preferred Genre: Sci-Fi
=====
User ID: 25
Movie Personality: The Action Hunter
Personality Insight: You like energetic, thrilling, and high-intensity movies.
Movie Twin: User 189
Taste Similarity: 46.42%
=====
Top Advanced Recommendations:

   title                                     genres  average_rating  rating_count  CineScore
7987  Star Wars: Episode VI - Return of the Jedi (1983)  Action|Adventure|Sci-Fi      4.14      196      92.33
7986  Star Wars: Episode V - The Empire Strikes Back...  Action|Adventure|Sci-Fi      4.22      211      89.13
5665  Minority Report (2002)  Action|Crime|Mystery|Sci-Fi|Thriller      3.64      120      82.54
6758  Prestige, The (2006)  Drama|Mystery|Sci-Fi|Thriller      4.01       90      82.07
4389  Interstellar (2014)  Sci-Fi|IMAX      3.99       73      80.48

Why these movies were recommended:
=====
Movie: Star Wars: Episode VI - Return of the Jedi (1983)
CineScore: 92.33/100
Reason:
- Matches your current mood: adventurous
- Belongs to genres: Action|Adventure|Sci-Fi
- Has an average rating of 4.14/5
- Rated by 196 users
- Also matches your preferred genre: Sci-Fi

```



8. Interface

An interactive interface was created using Gradio. The user can enter a User ID, select mood, choose preferred genre, and select the number of recommendations. The interface

displays personalized movie recommendations, Movie Personality, Movie Twin, CineScore, explanation, and weekend watchlist.

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory.

Enter User ID

20

Select Your Mood

romantic

Preferred Genre

Thriller

Number of Recommendations

5

Clear

Submit

CineMood AI Recommendations

Selected mood: romantic

Preferred Genre: Thriller

User ID: 20

Movie Personality: The Balanced Viewer

Personality Insight: You enjoy a mix of different movie genres.

Movie Twin: User 169

Taste Similarity: 32.18%

Top Recommendations:

Fight Club (1999)

Genres: Action|Crime|Drama|Thriller

Average Rating: 4.27/5

Rating Count: 218

CineScore: 88.14/100

Why recommended:

- Matches your mood: romantic

- Matches genre pattern: Action|Crime|Drama|Thriller

- Recommended using similar users' ratings

- Matches preferred genre: Thriller

Depicted, The (2006)

steps: [Deploy to Cloud Run](#)

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory.

CineMood AI

Personalized Movie Recommendation System based on Mood, Genre, Ratings & AI Similarity

Choose Your Movie Mood

User ID

29

Current Mood

happy

Preferred Genre

Drama

Your AI Movie Picks

CineMood AI Output

- Strong rating and popularity performance

- Matches your preferred genre: Drama

4. Life Is Beautiful (La Vita è bella) (1997)

Genres: Comedy|Drama|Romance|War

Average Rating: 4.15/5

Rated by: 88 users

CineScore: 84.4/100

Why recommended:

- Matches your happy mood

- Recommended using similar users' ratings

steps: [Deploy to Cloud Run](#)

9. Output

The system successfully recommends movies based on user rating patterns, mood, and genre preference. It also handles new users using cold-start recommendation based on movie popularity, average rating, mood, and genre.

10. Conclusion

CineMood AI successfully demonstrates a personalized movie recommendation system using collaborative filtering and cosine similarity. The project covers data preprocessing, exploratory data analysis, recommendation algorithm implementation, visualizations, and interactive interface development.

The system is more creative than a basic recommendation model because it includes mood-based filtering, Movie Twin, Movie Personality, CineScore, explainable recommendations, and a weekend watchlist. Overall, the project provides a simple, useful, and interactive AI-based movie recommendation solution.